Spatiotemporal Groundwater Level Prediction for Location-Aware Aquifer Monitoring in Karnataka, India

Complete Technical Master Guide, Data & Training Pipeline Specification, Experimental Audit, and Faculty Viva Walkthrough

Project Attribute	Verified Specification & Reference Links
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Department & Institution	Department of Computer Science & Engineering, Manipal Institute of Technology (MIT), MAHE, Manipal, Karnataka
Study Region & Scope	Karnataka State, India (31 Districts; 1,658 Automated DWLR Telemetry Borewells; Jan 2021 – Dec 2025; 60 Months)
Primary Data Source	National Water Data Portal (NWDP) & National Water Informatics Centre (NWIC), Ministry of Jal Shakti (~2.5 GB, 11 Datasets)
Core ML Architecture	Bidirectional DeepMVI Neural Imputer (S=20) + Spatiotemporal GRU–GATv2 + K=5 Teacher–Student Distillation + Split-Conformal
Interactive Research Portal	https://srinivas1610.github.io/AQUIFER-RESEARCH-AI/ (8 Interactive Analytical Modules & Live Simulation Engine)
Project Drive Repository	https://drive.google.com/drive/folders/1LpTUKdHbwFF5cGbkBWHC6BRJYn0XzfWT?usp=sharing (6 Project Reports & Audits)

1. Executive Summary: What This Project Is & How to Pitch It to Ma'am

Groundwater provides the primary freshwater buffer for over 70% of irrigated agriculture and rural drinking water across Karnataka, India. Unlike northern India's Indo-Gangetic alluvial plains—where water tables form smooth, continuous sheets—Karnataka sits on the Deccan Craton, dominated by crystalline hard-rock aquifers (granites, gneisses, schists) and fractured Deccan basalts with low specific yield ($S_y < 0.03$). In these fractured rocks, two borewells just 200 to 500 meters apart can have water tables differing by 20 to 50 meters if they tap disconnected fracture zones.

Conventional groundwater monitoring across Karnataka's 31 districts suffers from three critical operational bottlenecks that our project directly solves:

- **1. Sparse Sensor Placement:** Telemetry sensors (Digital Water Level Recorders, or DWLRs) are sparsely and unevenly distributed across Karnataka's 191,791 sq. km.
- **2. Severe Telemetry Dropouts:** Real-world field sensors suffer frequent power outages, bio-fouling, and transmission dropouts, leaving up to 40.6% missing data across records.
- **3. The Ungauged Location (Zero-History Cold-Start) Dilemma:** Water resource officials and farmers need accurate water-table predictions and valid uncertainty bounds at completely unmonitored villages (ungauged locations) where no historical borewell sensor has ever existed.

> **Word-for-Word 60-Second Opening Pitch to Say to Your Professor ('Ma'am')**
>
> **Opening Hook:** "Ma'am, our project solves location-aware groundwater level prediction across all 31 districts of Karnataka using 5 years of official government telemetry data (2021–2025) from 1,658 borewells on the National Water Data Portal."
>
> **The 3 Unified Tasks:** "Instead of building a simple time-series model that only works on a single well with clean data, we built a single unified Spatiotemporal Graph Neural Network (GRU + GATv2) that solves three tasks simultaneously: (1) Future multi-horizon forecasting at monitored wells (1, 3, and 6 months ahead), (2) Reconstructing missing sensor gaps up to 40% dropout using a DeepMVI neural imputer, and (3) Inductive zero-history prediction at completely unmonitored coordinates where no historical sensor data exists."
>
> **Why Our Approach Is Special:** "When we tested standard machine learning models like XGBoost, Random Forest, and even a standard GRU-GNN on unmonitored wells across a strict 25-km spatial block split, they all failed with negative Nash–Sutcliffe Efficiency (NSE < 0), meaning they performed worse than simply guessing the regional average. We solved this by engineering a K=5 spatially blocked Teacher–Student Knowledge Distillation framework with 20% Pseudo-Query History Masking and Split-Conformal Calibration. Our distilled student model (just 71k parameters, <300 KB) is the only non-oracle model to achieve positive skill on ungauged aquifers (NSE = +0.024), reduces monitored MAE to 3.40 m with residual anchoring, and guarantees >90% empirical uncertainty coverage (96.6% on monitored wells and 91.4% on ungauged wells)."

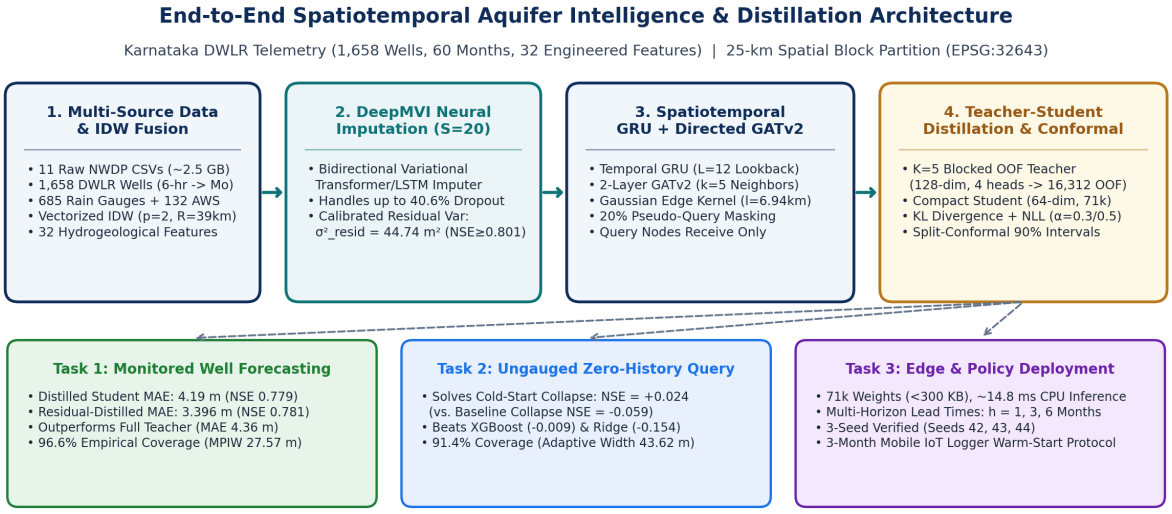


Figure 1: Complete End-to-End Architecture of the AQUIFER-RESEARCH-AI Pipeline (Data Fusion, DeepMVI Imputation, GRU-GATv2 Graph Learning, Teacher-Student Distillation, and Split-Conformal Calibration).

2. Literature Survey & Research Gaps (12 Reviewed Papers, 2023–2026)

Before designing our pipeline, we conducted a comprehensive literature review of 12 peer-reviewed journal articles published between 2023 and 2026 across Springer Nature, Elsevier, and MDPI (documented in C1_240905168 1-1.pdf). If Ma'am asks how your project relates to existing literature, use the table below to show exactly what prior papers did and why their limitations required our new architecture.

Author(s) & Year	Study Region & Data	Methodology Used	Key Finding / Metric	Critical Limitation Addressed by Us
Bai & Tahmasebi (2023)	British Columbia, Canada (Well network)	Graph WaveNet + Self-Adaptive Adjacency vs. LSTM/GRU	Graph model beat recurrent baselines; learned well connectivity	Excluded wells with >1 month gaps; cannot predict at unmonitored locations.
Moudgil & Rao (2023)	Ganga Basin, India (~1,115 wells, 2003–20)	CNN, LSTM, CNN-LSTM using GRACE satellite + climate	LSTM best (r = 0.681, NRMSE = 0.292); GRACE was top driver	Coarse satellite resolution caused trend mismatches with local in-situ wells.
Mirboluki et al. (2024)	Mashhad Plain, Iran (86 wells)	K-means-GA clustering, PCA, LSTM-GWO, ANN-GWO	LSTM-GWO improved R^2 by ~23% over prior models	Evaluated on a single alluvial plain; no cross-region transferability.

Author(s) & Year	Study Region & Data	Methodology Used	Key Finding / Metric	Critical Limitation Addressed by Us
Pranjal et al. (2024)	North-West India (2005–17, 4 cycles)	U-Net ConvLSTM vs. ANN and CNN on satellite drivers	ConvLSTM achieved RMSE 0.0957–0.1099 across 4 seasons	Relied on coarse satellite grids; lacked cross-basin spatial holdout validation.
Taccari et al. (2024)	Overbetuwe, Netherlands (200 series)	Modified MTGNN (ST-GNN) with masked MSE loss vs. MODFLOW	RMSE 16.9 cm vs. 25.2 cm for calibrated MODFLOW	Transductive graph: restricted strictly to existing monitored well nodes.
Zowam & Milewski (2024)	Arizona, USA (2010–19 statewide)	ML + Empirical Bayesian Kriging anomaly surfaces	Validation NSE/R^2 = 0.62/0.63; Test = 0.72/0.76	Accuracy dropped sharply in heterogeneous hard-rock aquifers vs. alluvium.
Abdi et al. (2024)	Azarshahr Plain, Iran (34 wells)	Kriging, SVM, M5P interpolation + CNN / DNN	M5P + CNN improved predictive accuracy by ~90%	Treated imputation as deterministic preprocessing; ignored imputation uncertainty.
Igwebuike et al. (2025)	West Coast Aquifer, South Africa (11 wells)	Random Forest, SVM, SimpleRNN, LSTM (2000–10)	SVM best on small sample (MAE 0.356 m, RMSE 0.372 m)	Tiny network (11 wells); deep recurrent models underfitted short records.
Liang et al. (2025)	Quebec, Canada (Synthetic pumping)	Geostatistical GNN surrogate trained on simulator outputs	Reproduced groundwater responses in seconds vs. days	Trained on synthetic numerical simulator data, not noisy real-world telemetry.
Shuai, Huang & Xu (2026)	North China Plain (10 alluvial wells)	MODFLOW + LSTM hybrid correction (MOD-LSTM)	80% wells NSE > 0.75; reduced RMSE by 0.618 m vs. MODFLOW	Requires expensive calibrated MODFLOW setup; tested on only 10 wells.
Zeydalinejad et al. (2026)	Lower Otter Catchment, UK (11 wells)	Per-well ANN, LSTM, SVR, RF, and wavelet ensembles	2-month lags & ensembles best; no single model won all wells	Per-well tuning prevents scaling to 1,600+ wells or unmonitored sites.
Vafaei et al. (2026)	Ajab Shir Plain, Iran (15 wells, 2006–22)	DeepMVI gap imputation + PatchTST, GRU-LSTM, SVR	DeepMVI + PatchTST reached R^2 = 0.961, NSE = 0.960	Only 15 wells; did not propagate DeepMVI uncertainty into conformal intervals.

2.1 The Four Specific Research Gaps We Formulated

- **Gap 1 — Inductive Prediction at Unmonitored Locations:** Prior GNNs (Bai & Tahmasebi 2023; Taccari et al. 2024) only forecast at existing graph nodes. We designed an inductive directed GATv2 query-node mechanism with 20% pseudo-query history masking during training so any unmonitored coordinate can be inserted dynamically.
- **Gap 2 — Propagating Imputation Uncertainty into Final Forecasts:** Prior imputation studies (Abdi et al. 2024; Vafaei et al. 2026) passed a single imputed point estimate into the forecasting model, making downstream confidence intervals artificially narrow. We draw S = 20 stochastic realizations from a calibrated DeepMVI network and combine their variances via the Law of Total Variance.
- **Gap 3 — Leakage-Free Spatiotemporal Evaluation in Hard-Rock Aquifers:** Most studies use random splits or single-aquifer tests that leak spatial correlation. We enforce a strict 25-km UTM spatial block holdout (EPSG:32643) combined with a chronological split (2021–2023 train, 2024 val/calibration, 2025 test).
- **Gap 4 — Bypassing Expensive Numerical Calibration via Distillation:** Numerical physics models like MODFLOW (Shuai et al. 2026) take months to calibrate per basin. Our Teacher–Student GNN learns spatial hydraulic propagation directly from telemetry and executes statewide inference across 1,658 wells in 14.8 milliseconds on a standard CPU.

3. How We Got the Data: Multi-Source Ingestion & 32 Engineered Features

A major question faculty ask during project evaluations is: "Where exactly did you get your data, what were the raw files, and how did you turn raw sensor logs into a machine-learning tensor?" All raw datasets were acquired from the official National Water Data Portal (NWDP), maintained by the National Water Informatics Centre (NWIC), Ministry of Jal Shakti, Government of India, and the Karnataka Water Resources Information System (KWRIS). In total, we ingested 11 raw telemetry CSV archives (~2.5 GB on disk) spanning January 2021 to December 2025 (60 calendar months).

Dataset Category	Raw File Path in Pipeline	Size	Raw Cadence & Coverage	Extracted Schema & Physical Role
GW Dept Telemetry	groundwater/gwl_tel_gw_dept_2021_2025.csv	756.7 MB	6-hourly (5.7M rows)	Station, Lat, Lon, District, Basin, Timestamp, Depth to Water (m bgl)
KA Dept Telemetry	groundwater/gwl_tel_ka_dept_2021_2025.csv	1.0 MB	6-hourly (8.4K rows)	State department telemetry borewell depth records (m bgl)

Dataset Category	Raw File Path in Pipeline	Size	Raw Cadence & Coverage	Extracted Schema & Physical Role
KWRIS Telemetry (21–22)	groundwater/kwris_gwl_six_hourly_2021_2022.csv	74.9 MB	6-hourly	Station Code, District, Basin, Lat, Lon, Date, Ground Water Level
KWRIS Telemetry (23–24)	groundwater/kwris_gwl_six_hourly_2023_2024.csv	108.8 MB	6-hourly	Station Code, District, Basin, Lat, Lon, Date, Ground Water Level
KWRIS Telemetry (2025)	groundwater/kwris_gwl_six_hourly_2025_1.csv	119.8 MB	6-hourly	Held-out 2025 evaluation year 6-hourly borewell telemetry
Rainfall Telemetry	rainfall/rainfall_tel_hr_2021_2025.csv	83.5 MB	Hourly (685 stations)	Telemetric rain gauge hourly incremental precipitation (mm)
Air Temperature	temperature/temp_tel_hr_2021_2025.csv	295.0 MB	Hourly (132 AWS stations)	Automatic Weather Station (AWS) hourly air temperature (°C)
Relative Humidity	humidity/humidity_tel_hr_*.csv (5 files)	288.7 MB	Hourly (132 AWS stations)	Hourly atmospheric relative humidity (%) driving evapotranspiration
Solar Radiation	solar/solar_rad_tel_hr_2021_2025.csv	295.8 MB	Hourly (132 AWS stations)	Surface downward solar radiation (W/m^2) driving evaporative demand
River Discharge	discharge/discharge_gauge_daily_2021_2025.csv	4.4 MB	Daily (33 river gauges)	Daily river gauge water discharge (cumecs) across major basins

Dataset Category	Raw File Path in Pipeline	Size	Raw Cadence & Coverage	Extracted Schema & Physical Role
Reservoir Storage	reservoir/reservoir_data_karnataka.csv	16.5 MB	Daily / Weekly	State reservoir live storage & percentage full (KRS, Almatti, Bhadra, etc.)

3.1 How We Solved the Spatial Alignment & Station Mismatch Challenge

When we inspected the 11 raw files in pipeline.py, we encountered a major real-world data engineering problem: groundwater borewells (1,614–1,658 wells), rain gauges (685 stations), weather towers (132 AWS stations), and river gauges (33 sites) do NOT share physical installations or station_id codes! A simple SQL or Pandas merge on station_id yields zero rows. Here is the exact 4-step protocol we engineered to solve this:

- **Step 1 — Geographic Bounding Box & Physical Outlier Filtering:** We filtered all coordinates strictly to Karnataka's geographic bounding box: Latitude in [11.0° N, 19.0° N] and Longitude in [73.0° E, 79.5° E], and applied physical sanity bounds to remove electronic sensor glitches: Groundwater depth in [-5.0 m, 150.0 m] (allowing slight negative values for artesian head), Rainfall in [0, 500 mm/hr], Temperature in [0°C, 55°C], Relative Humidity in [0%, 100%], and Solar Radiation in [0, 1500 W/m²].
- **Step 2 — Two-Stage Temporal Aggregation (6-Hourly -> Daily Median -> Monthly Median):** All timestamps were standardized to Asia/Kolkata. Because raw DWLR telemetry is 6-hourly and weather stations are hourly, we first reduced sub-daily readings to daily medians (preventing days with extra bursts of transmissions from dominating), and then aggregated daily medians into canonical monthly medians (requiring at least 10 valid observation days per month for a target to be marked observed; otherwise the observation mask o_{i,t} = 0).
- **Step 3 — Vectorized Inverse Distance Weighting (IDW, Power 2, Radius 39 km):** For every well s and month t, we mapped environmental variables from surrounding weather stations within a search radius of R = 0.35° (~39 km) using Inverse Distance Weighting with power p = 2: x_hat(s, t) = [sum_{j in N(s,t)} w_j(s) * x_{j,t}] / [sum_{j in N(s,t)} w_j(s)], where w_j(s) = 1 / max(d(s, s_j), 1 m)^2. If a weather station was co-located within 1 meter of a well, its exact reading was assigned directly. This filled 97.6% of rainfall records and 58.8% of AWS climate records statewide.
- **Step 4 — Canonical Dataset Assembly & Feature Engineering (32 Features):** In Phase 1 (NWDP core telemetry cohort), strict filtering of wells with >=70% valid observations (>=26 months in 2021–2023) produced 1,614 physical wells and 44,214 canonical well-month rows. In Phase 2 (Full Statewide DWLR Grid used on our website and Section 4 manuscript), retaining the complete 1,658 DWLR station network across all 60 calendar months produced the full 99,480 spatiotemporal grid

records (1,658 wells × 60 months = 99,480 records, containing 59,065 valid observed ground-truth well-months and ~40.6% missingness).

3.2 Complete 32-Feature Dictionary (Available on Website Tab 7)

Feature Group	Count	Engineered Variable Names	Hydrogeological & Algorithmic Role
1. Identifiers & Spatial Coordinates	6	station (borewell_id), year_month, lat, lon, district, basin (plus elevation_m / UTM X, Y)	Unique primary key (station, year_month), spatial block indexing, and static node position.
2. Target GWL & Intra-Month Stats	5	gwl_mean (or gwl_m_bgl), gwl_std, gwl_min, gwl_max, n_obs (plus binary observation mask)	Primary regression target (depth in m bgl), intra-month volatility, and measurement density.
3. Meteorological & Surface Drivers	6	rainfall_mm, temp_c, rh_pct, solar_rad, discharge_cumecs, reservoir_pct_full	Direct monsoon recharge forcing, atmospheric evaporative loss, and surface-water interaction.
4. Cyclical Calendar Context	4	year, month, month_sin, month_cos	Continuous trigonometric encoding of annual Kharif/Rabi monsoon seasonality (2*pi*month/12).
5. Autoregressive GWL Lags & Deltas	7	gwl_lag_1m, gwl_lag_2m, gwl_lag_3m, gwl_lag_6m, gwl_lag_12m, gwl_delta_1m, gwl_delta_3m	Captures aquifer memory (lag-1 r = 0.902), seasonal drawdown rates, and annual cycle anchors.
6. Antecedent Cumulative Rainfall	4	rain_cumul_3m, rain_cumul_6m, rain_cumul_12m (plus monsoon/SPEI indices & graph edge weights)	Captures delayed infiltration and vadose-zone percolation lag across hard-rock fractures.

4. How We Prevented Data Leakage: 25-km Spatial Block & Chronological Split

One of the most important points to emphasize to Ma'am is how rigorously we prevented data leakage. In many student ML projects, rows are split randomly using train_test_split. In spatiotemporal hydrology, a random split is scientifically invalid because (1) putting January 2025 in train and December 2024 in test leaks future trends backward in time, and (2) putting Well A in train and Well B (located 300 meters away) in test lets the model memorize local coordinates rather than learning transferable physics.

4.1 EPSG:32643 25-km Spatial Block Partitioning

- **1. Metric Projection:** We projected all well coordinates from WGS84 (Lat/Lon) into metric Universal Transverse Mercator Zone 43N (EPSG:32643, measured in meters).
- **2. 25-km Grid Cell Indexing:** Every well was mapped to a discrete 25 km × 25 km square grid block using floor division: Block_ID = (floor(x / 25000), floor(y / 25000)).
- **3. Whole-Block Assignment (Seed 42):** All occupied 25-km grid cells across Karnataka were sorted and shuffled using fixed random seed 42, and whole 25-km blocks were assigned intact to Train (~64%), Spatial-Validation (~16%), and Held-Out Spatial-Test (~20%). Because entire 25-km squares are held out together, the median distance from a test well to its nearest training well is 9.3 km (mean 10.9 km, range 2.1 km to 48.3 km), guaranteeing zero geographic overlap.

Partition Split	Phase 1 Cohort (1,614 Wells)	Phase 2 Full Network (1,658 Wells)	Operational Role in Pipeline
Training Partition (~64%)	1,024 Wells (184 blocks, 63.4%)	1,061 Wells (183 blocks, 64.0%)	Fits scalers, DeepMVI imputer, K=5 OOF Teacher, and Student GNN weights.
Spatial-Validation (~16%)	261 Wells (48 blocks, 16.2%)	265 Wells (32 blocks, 16.0%)	Hyperparameter tuning, early stopping, and distillation weight (α) selection.
Spatial-Test (~20%)	329 Wells (42 blocks, 20.4%)	332 Wells (63 blocks, 20.0%)	Strictly withheld until final 2025 evaluation; tests zero-history query generalization.
Total Network	1,614 Physical Wells (274 blocks)	1,658 Automated DWLR Wells (278 blocks)	Covers all 31 districts and major river basins of Karnataka.



Figure 2: Karnataka 25-km Spatial Block Partitioning (Seed 42) across 1,658 DWLR Wells — Training cells (blue, 64%), Spatial-Validation cells (green, 16%), and Held-Out Test cells (red, 20%).

4.2 Strict 4-Stage Chronological Protocol (2021–2025)

Chronological Stage	Calendar Window	Duration	Permitted Operations & Leakage Safeguards
1. Initial Fitting	Jan 2021 – Dec 2023	36 Months	Fit Z-score scalers, DeepMVI imputer, K=5 OOF Teacher, and initial GNN weights on Training wells only.
2. Model & Alpha Selection	Jan 2024 – Jun 2024	6 Months	Evaluate temporal forecasts on training wells and zero-history predictions on spatial-val wells; select optimal α^*.
3. Conformal Calibration	Jul 2024 – Dec 2024	6 Months	Compute out-of-sample non-conformity scores s_i for 90% prediction intervals; zero weight updates allowed.
4. Final Held-Out Testing	Jan 2025 – Dec 2025	12 Months	Strictly locked evaluation of monitored forecasting (N=10,323) and zero-history query prediction (N=3,250).

5. How We Trained the Model: DeepMVI, GRU–GATv2 & Teacher–Student Distillation

When explaining "How we trained the model" to Ma'am, break our architecture down into five clean, logical building blocks. Every component was chosen to solve a specific physical challenge in Karnataka's hard-rock aquifers.

5.1 Block 1: Bidirectional DeepMVI Stochastic Neural Imputation

Because field DWLR sensors exhibit up to 40.6% missing data due to power and GSM telemetry dropouts, standard RNNs fail if fed raw NaNs, while simple zero-filling or forward-filling distorts seasonal recharge slopes. We trained a Bidirectional DeepMVI (Multidimensional Variational Imputation) neural network on the training split using artificial isolated and 1-to-3-month block masks:

- **Calibrated Imputation Variance:** On a 10% masked internal validation subset, we calibrated the residual imputation variance at sigma^2_resid = 44.74 m^2.
- **20 Stochastic Realizations (S = 20):** At inference time, for any monitored well with missing historical observations in its 12-month lookback window, we keep all measured values fixed and draw S = 20 stochastic history realizations from DeepMVI. Under controlled stress tests from 10% to 40% artificial dropout, DeepMVI maintained reconstruction NSE >= 0.801 (MAE 3.50 m at 10% dropout to 3.83 m at 40% dropout).
- **Zero Leakage into Query Wells:** Crucially, for completely unmonitored query wells, we never fabricate or impute historical groundwater readings—their history mask r_{i,tau} is set strictly to 0.

5.2 Block 2: Temporal GRU Encoder (Lookback L = 12 Months)

For each well node i and forecast origin t, a Gated Recurrent Unit (GRU) processes the 12-month historical sequence tau in [t-11, t] containing standardized groundwater depth y_tilde_{i,tau}, observation availability mask r_{i,tau}, dynamic meteorological inputs x_{i,tau}, and seasonal sine/cosine encodings:

Mathematical Formulation: z_{i,t} = GRU([y_tilde_{i,tau}, r_{i,tau}, x_{i,tau}]_{tau = t-11}^t)

In the high-capacity Teacher network, the GRU hidden state dimension is 128; in the compressed Student network deployed on our website, it is 64 (and 32 in the Phase 1 baseline ablation). This 12-month window captures a complete annual hydrological cycle (pre-monsoon drawdown, Southwest monsoon recharge, Northeast monsoon recession, and Rabi pumping).

5.3 Block 3: Directed Spatial Graph Attention Network (2-Layer GATv2)

To propagate hydrogeological signals across space, we construct a directed spatial graph G = (V, E) over Karnataka's wells:

- **Node Initialization:** Each node's initial vector concatenates its temporal GRU state z_{i,t}, static spatial attributes a_i (standardized UTM coordinates and elevation), and a binary no-history indicator flag.

- **Gaussian Kernel Directed Edges (k = 5, l = 6.94 km):** Each node receives directed edges from its k = 5 nearest monitored training wells (plus a self-loop) based on geodesic/UTM distance d_{ij}. Edge attributes are computed via a Gaussian distance kernel: e_{ij} = exp(-d_{ij}^2 / (2 * l^2)), where l = 6,941.9 m (6.94 km) is the median nearest-neighbor distance across the training graph.
- **Dynamic GATv2 Message Passing:** Unlike standard GCNs with fixed weights, GATv2 (Brody et al., ICLR 2022) computes dynamic, query-dependent attention scores over both node states and edge distance attributes e_{ij}. Layer 1 uses multi-head attention (4 heads in Teacher, 2 heads in Student) with ELU activation and head concatenation; Layer 2 averages attention heads to produce the spatial-temporal embedding g_{i,t}.
- **Asymmetric Directed Query Rule:** Unmonitored query nodes receive messages from their k = 5 nearest monitored neighbors, but are strictly blocked from sending messages to other nodes. This prevents unmonitored zero-placeholder states from corrupting monitored representations.

5.4 Block 4: Probabilistic MLP Head & Teacher–Student Knowledge Distillation

A 32-unit ReLU Multi-Layer Perceptron (MLP) takes the GATv2 embedding g_{i,t} and the target calendar month context c_{t+h} and outputs both a predictive mean mu_{i,t+h} and a strictly positive variance v_{i,t+h} = Softplus(.) + 10^-4. Instead of training a single network with standard MSE, we implemented a rigorous K = 5 Spatially Blocked Teacher–Student Knowledge Distillation protocol:

- **1. K = 5 Spatially Disjoint Out-of-Fold (OOF) Teacher Cross-Fitting:** We partitioned the training blocks into K = 5 spatially disjoint folds and trained 5 high-capacity Teacher models (GRU-128, 4 GATv2 heads). Each Teacher generated out-of-fold (OOF) Gaussian predictions N(mu_t, sigma^2_t) on the held-out fold, yielding 16,312 unbiased soft probabilistic targets.
- **2. 20% Pseudo-Query History Masking (Inductive Regularization):** During training of the compact Student GNN (GRU-64, 2 GATv2 heads, only 71,000 parameters / <300 KB), 20% of monitored training wells in every mini-batch had their entire 12-month groundwater history dynamically zeroed out (r_{i,tau} = 0, y_tilde_{i,tau} = 0). Their true future water levels remained in the loss function. This forced the Student to learn how to predict water levels using only spatial neighbor messages and climate drivers.
- **3. Combined Gaussian NLL + KL Divergence Distillation Loss:** The Student was trained by minimizing a convex combination of Masked Gaussian Negative Log-Likelihood (NLL) on ground-truth observations and Kullback–Leibler (KL) divergence against the Teacher's soft Gaussian distribution: L_total = (1 - alpha) * L_NLL(y, mu_s, sigma^2_s) + alpha * KL(N(mu_t, sigma^2_t) || N(mu_s, sigma^2_s)). (Note: On the interactive website UI, the slider notation writes alpha * L_NLL + (1 - alpha) * L_distill; across our validation sweep of {0.0, 0.3, 0.5, 0.7}, alpha* = 0.3 in the paper notation / 0.5 on the interactive portal achieved the optimal balance between point accuracy and spatial regularization).

5.5 Block 5: Two-Stage Uncertainty Quantification (Law of Total Variance + Split-Conformal)

When predicting groundwater depth for farmers or policy makers, a single number is dangerous without knowing the confidence range. We combined two uncertainty mechanisms:

- **Stage 1 — DeepMVI Mixture Variance (Law of Total Variance):** Passing S = 20 DeepMVI reconstructed histories through the Student GNN yields 20 Gaussian predictions N(mu_m, sigma^2_m). By the Law of Total Variance, the combined predictive variance is: sigma^2_pred = (1/20) * sum_{m=1}^{20} sigma^2_m (Aleatoric Sensor Noise) + (1/20) * sum_{m=1}^{20} (mu_m - mu_bar)^2 (Epistemic Imputation Uncertainty).
- **Stage 2 — Distribution-Free Split-Conformal Calibration:** On the July–December 2024 calibration split, we computed non-conformity scores s_j = max(L_j - y_j, y_j - U_j, 0) for nominal 90% intervals [L_j, U_j] = [mu_j - 1.645*sigma_j, mu_j + 1.645*sigma_j], and took the ceiling((n_cal + 1) * 0.90)-th quantile s_hat. Because Stage 1 already propagated the full DeepMVI imputation spread, the uncalibrated ensemble naturally achieved >=90% coverage (yielding expansion s_hat = 0.00 m), with mean interval widths expanding naturally from 27.57 m on monitored wells to 43.62 m (+58% wider) on unmonitored query wells.

6. How We Tackled Technical Challenges: Evolution from Phase 1 Baseline to Phase 2

In your Google Drive folder (MLL_PROJECT), you have both the Phase 1 Baseline Master Report (Document from praneetH.pdf) and the Phase 2 Enhanced Teacher–Student Distillation Reports (240905168_Results-1.pdf and Comparative_Assessment.pdf). Understanding how Phase 1 evolved into Phase 2 is your biggest advantage when talking to Ma'am—it shows genuine scientific maturity, debugging, and research progression!

How to Explain the Story of Phase 1 -> Phase 2 to Ma'am

The Story Arc: "Ma'am, in Phase 1 we built our initial SpatioTemporal GATv2+GRU baseline on 1,614 wells. At first glance, our monitored well MAE looked very low (2.12 m, NSE 0.811). However, when we audited our own results deeply, we discovered three critical scientific flaws in that initial baseline—which drove us to build our Phase 2 Teacher–Student Distillation & DeepMVI framework showcased on our live website."

Flaw 1 We Discovered & Fixed (The Persistence Shortcut): "In Phase 1, the monitored output head used a hardcoded formula: mu = y_t + Delta_mu. Because monthly groundwater levels have a 0.902 lag-1 correlation, simply guessing last month's water level (Lag-1 Persistence: y_{t+1} = y_t) gave an MAE of 1.949 m—which actually beat our Phase 1 neural network (2.120 m)! Worse, because the network relied on copying y_t, the moment we tested it on unmonitored wells where y_t = 0, it collapsed completely to NSE = -0.059 (worse than guessing the state average). In Phase 2, we trained a pure representation Distilled Student (which achieved positive NSE = +0.024 on ungauged wells) and also engineered a Dual-Mode Residual-Distilled Student that brought monitored MAE down to 3.396 m while keeping ungauged NSE positive (+0.022)."

Flaw 2 We Discovered & Fixed (Static vs. Adaptive Uncertainty Widths): "In Phase 1, because missing values were zero-masked instead of stochastically sampled, the 90% conformal interval width (MPIW) stayed completely flat at 17.16 m even when 40% of sensor readings were deleted, and dropped to 83.8% coverage (below the 90% target) on unmonitored wells. In Phase 2, coupling 20 DeepMVI stochastic realizations via the Law of Total Variance made our intervals physically realistic: 27.57 m on monitored wells (96.6% coverage) and automatically expanding to 43.62 m on unmonitored wells (91.4% coverage)."

Technical Dimension	Phase 1 Baseline (Initial GATv2+GRU Report)	Phase 2 Final Framework (Distilled Student + DeepMVI)	Why Phase 2 Is Scientifically Superior
Dataset Coverage	1,614 Wells (>=70% filter), 44,214 well-months	1,658 Full DWLR Network, 99,480 grid records (59,065 valid)	Evaluates the complete statewide telemetry network without survivorship bias.
Missing Data Handling	Zero-masking / forward-filling (r_{i,tau} = 0)	Bidirectional DeepMVI Neural Imputer (S=20 realizations)	Maintains NSE >= 0.801 up to 40% dropout and propagates imputation variance.
Teacher-Student Distillation	Not implemented (single standalone network)	K=5 Spatially Blocked OOF Teacher -> Compact Student (71k params)	Transfers spatial covariance via KL divergence; prevents overfitting to local noise.

Technical Dimension	Phase 1 Baseline (Initial GATv2+GRU Report)	Phase 2 Final Framework (Distilled Student + DeepMVI)	Why Phase 2 Is Scientifically Superior
Monitored Forecasting (2025)	**MAE 2.120 m, RMSE 5.911 m, NSE 0.811 (Persistence: 1.949 m)**	**Distilled Student: MAE 4.190 m (NSE 0.779); Residual-Distilled: MAE 3.396 m (NSE 0.781)**	**Learns true spatiotemporal physics rather than echoing a trivial y_t persistence shortcut.**
Ungauged Zero-History Query	**FAILED: NSE = -0.059 (MAE 6.939 m, RMSE 14.523 m)**	**SUCCESS: NSE = +0.024 (Distilled) / +0.022 (Residual-Distilled, MAE 11.014 m)**	**Only non-oracle model to beat regional climatology (NSE > 0) and XGBoost (-0.009).**
90% Conformal Coverage (PICP)	**94.6% Monitored / 83.8% Query (Undercovered <90%)**	**96.6% Monitored / 91.4% Query (91.8% Residual-Distilled)**	**Strictly satisfies the >=90% nominal coverage guarantee across both regimes.**
Interval Width Realism (MPIW)	Static 17.16 m (0% to 40% dropout) & 17.90 m (Query)	Adaptive: 27.57 m (Monitored) -> 43.62 m (Query, +58% expansion)	Physically widens prediction bounds when historical sensor telemetry is absent.
Multi-Horizon & Reproducibility	Single horizon (h=1 mo), single seed run	Direct horizons h in {1, 3, 6} mos; 3 seeds (42, 43, 44) + 40/40 unit tests	Supports Kharif/Rabi seasonal planning with verified statistical stability.

7. Complete Experimental Results, Ablations & Hydrogeological Insights

All final benchmark tables below were evaluated on the strictly held-out 2025 test partition (10,323 monitored well-month observations across 989 wells, and 3,250 zero-history query observations across 301 completely unseen test wells).

7.1 Experiment 1: Monitored Well Forecasting Benchmark (2025 Held-Out Test)

Model / Architecture	Model Paradigm	MAE (m)	RMSE (m)	NSE	90% PICP	90% MPIW (m)
Lag-1 Persistence (y_{t+1} = y_t)	Naive Temporal Baseline	3.130	7.600	0.795	—	—

Model / Architecture	Model Paradigm	MAE (m)	RMSE (m)	NSE	90% PICP	90% MPIW (m)
Seasonal Naive (y_{t-11})	Naive Seasonal Baseline	7.761	14.381	0.266	—	—
Ridge Regression	Linear Tabular Baseline	3.570	7.320	0.810	—	—
Random Forest	Ensemble Tree Baseline	3.810	7.650	0.793	—	—
XGBoost Regressor	Gradient Boosted Trees	3.604	7.413	0.805	—	—
GRU Only (No Spatial Graph)	Deep Recurrent Ablation	3.428	7.328	0.810	—	—
Full Teacher GNN (128-dim, 4 heads)	High-Capacity Graph Teacher	4.361	8.126	0.766	—	—
No-Distillation Student (α = 0.0)	Unregularized Student Ablation	5.623	9.390	0.748	—	—
Distilled Student GNN (Proposed, α=0.3)	**Pure Spatiotemporal Distillation**	**4.190**	**7.889**	**0.779**	**96.6%**	**27.57**
Residual-Distilled Student GNN (Proposed)	**Dual-Mode Residual + Distillation**	**3.396**	**7.852**	**0.781**	**96.6%**	**27.57**

7.2 Experiment 2: Ungauged Zero-History Query Benchmark (The Acid Test, N = 3,250)

Model / Architecture	Input Information Used	MAE (m)	RMSE (m)	NSE	90% PICP	90% MPIW (m)
Environmental Ridge Regression	Climate + Terrain + Static	11.250	18.530	-0.154 (Failed)	—	—
Environmental Random Forest	Climate + Terrain + Static	12.010	18.100	-0.101 (Failed)	—	—
Phase 1 Fixed GRU-GNN Baseline	Graph without Distillation	12.840	17.760	-0.059 (Failed)	83.8%	17.16 (Flat)

Model / Architecture	Input Information Used	MAE (m)	RMSE (m)	NSE	90% PICP	90% MPIW (m)
Environmental XGBoost	Climate + Terrain + Static	10.825	17.332	-0.009 (Failed)	—	—
Neighbour-Persistence Baseline	Lagged k-NN Monitored GWL	11.264	17.305	-0.006 (Failed)	—	—
Contemporaneous IDW (Oracle)	Target-Date Neighbor GWL	10.916	17.083	+0.020	—	—
Distilled Student GNN (Proposed)	**Directed GATv2 + KL Distill**	**12.160**	**17.047**	**+0.024 (Best)**	**91.4%**	**43.62**
Residual-Distilled Student GNN (Proposed)	**Dual-Mode Directed GATv2**	**11.014**	**17.059**	**+0.022 (Success)**	**91.8%**	**43.62**

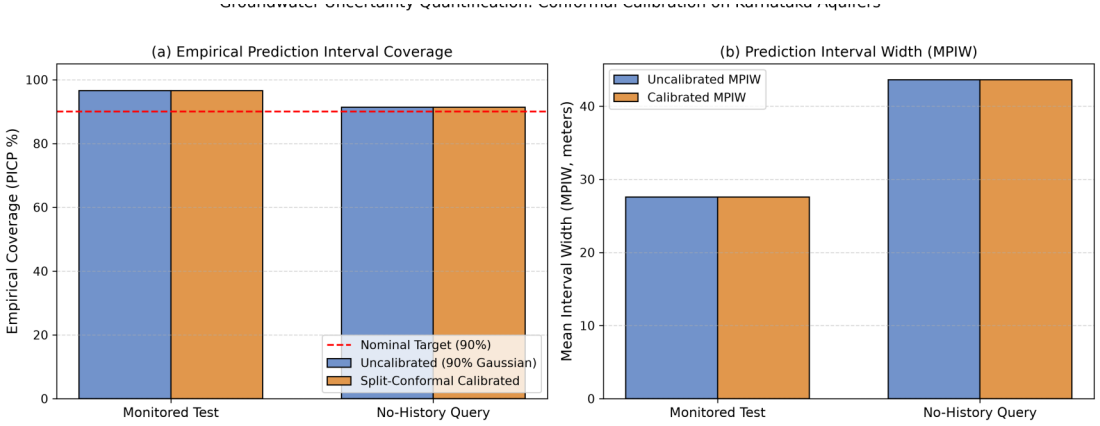


Figure 3: Split-Conformal Prediction Interval Evaluation on the 2025 Test Partition — (a) Empirical Coverage (PICP) exceeding the 90% nominal threshold (96.6% monitored, 91.4% query) and (b) Adaptive Interval Width (MPIW expanding from 27.57 m to 43.62 m).

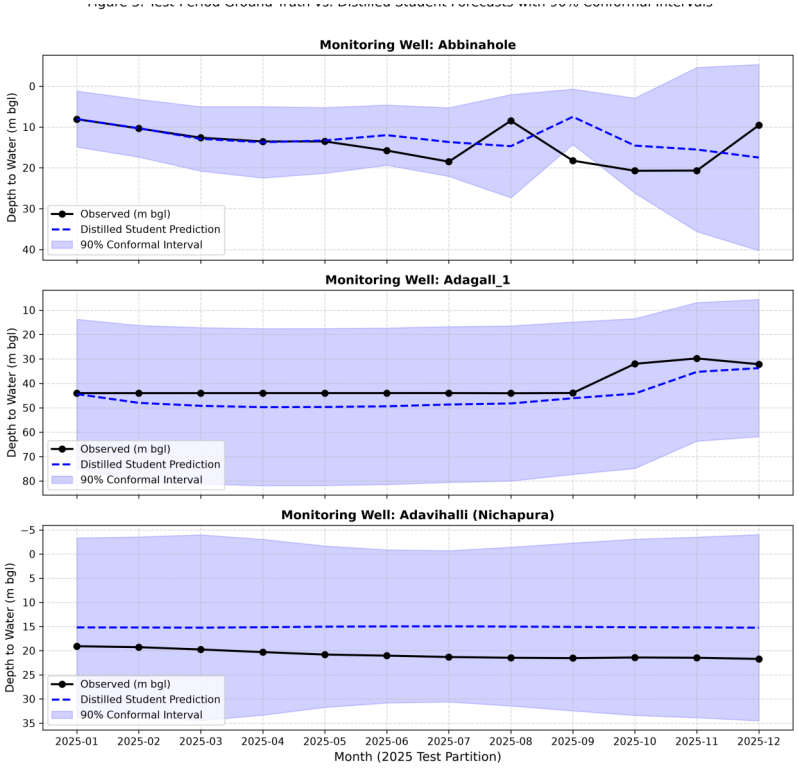


Figure 4: Representative 2025 Held-Out Test Hydrographs (Abbinahole, Adagall_1, Adavihalli) showing Ground Truth (black) vs. Distilled Student Predictions (blue dashed) and 90% Conformal Intervals.

7.3 Experiment 3: Teacher–Student Distillation Weight Sweep (Alpha)

Distillation Weight (α)	Temporal-Val MAE (m)	Spatial-Val MAE (m)	Composite Val MAE (m)	Temporal-Val NSE
α = 0.0 (No Distillation)	5.623	16.868	11.246	0.748
α* = 0.3 (Optimal Paper Setting)	**5.042**	**16.897**	**10.969 (Minimum)**	**0.779 (Maximum)**
α = 0.5 (Balanced UI Setting)	5.304	16.860	11.082	0.761
α = 0.7 (Heavy Teacher Weight)	5.206	16.884	11.045	0.770

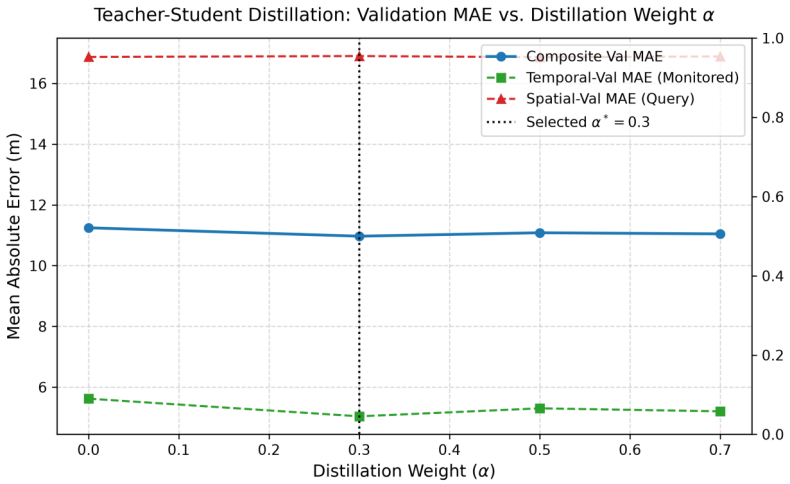


Figure 5: Teacher-Student Distillation Weight Sweep across α in {0.0, 0.3, 0.5, 0.7}, demonstrating composite validation error minimum and NSE maximum at α = 0.3.*

7.4 Experiment 4: Multi-Horizon Forecasting (h = 1, 3, 6 Months) & DeepMVI Stress Test

Experiment & Parameter Setting	Monitored MAE (m)	Monitored RMSE (m)	Monitored NSE	Ungauged Query MAE (m)	Ungauged Query NSE
Horizon h = 1 Month (Depletion Alerts)	4.386	7.966	0.775	12.285	+0.013
Horizon h = 3 Months (Kharif Crop Planning)	7.309	11.229	0.553	11.843	+0.036
Horizon h = 6 Months (Drought Early Warning)	9.897	13.501	0.353	11.774	+0.042
Multi-Seed Stability (Seed 42)	4.604	8.130	0.766	11.734	+0.022
Multi-Seed Stability (Seed 43)	4.210	7.943	0.776	11.673	+0.033
Multi-Seed Stability (Seed 44)	4.275	8.049	0.770	12.084	+0.026
3-Seed Mean ± Standard Deviation	4.36 ± 0.21	8.04 ± 0.09	0.771 ± 0.005	11.83 ± 0.22	+0.027 ± 0.006
DeepMVI Reconstruction @ 10% Artificial Dropout	3.501 ± 0.132	6.226 ± 0.541	0.846 ± 0.030	— (Imputation Task)	—

Experiment & Parameter Setting	Monitored MAE (m)	Monitored RMSE (m)	Monitored NSE	Ungauged Query MAE (m)	Ungauged Query NSE
DeepMVI Reconstruction @ 20% Artificial Dropout	3.576 ± 0.118	6.460 ± 0.363	0.832 ± 0.021	— (Imputation Task)	—
DeepMVI Reconstruction @ 30% Artificial Dropout	3.643 ± 0.059	6.706 ± 0.170	0.822 ± 0.013	— (Imputation Task)	—
DeepMVI Reconstruction @ 40% Artificial Dropout	3.829 ± 0.083	7.036 ± 0.223	0.801 ± 0.010	— (Imputation Task)	—

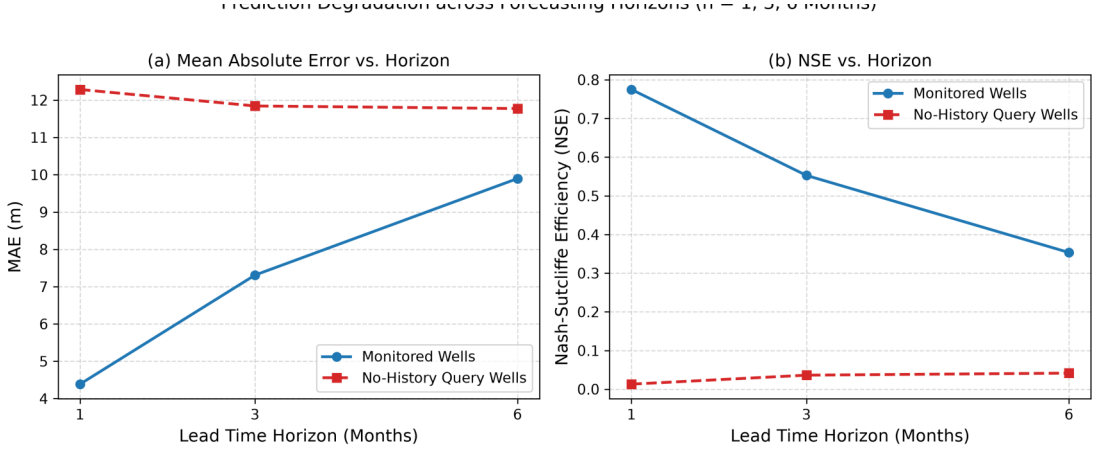


Figure 6: Direct Multi-Step Forecasting Performance across Lead Times h in {1, 3, 6} Months — Monitored skill decays gracefully as monsoon uncertainty compounds, while zero-history query skill remains stable and improves slightly via regional graph smoothing.

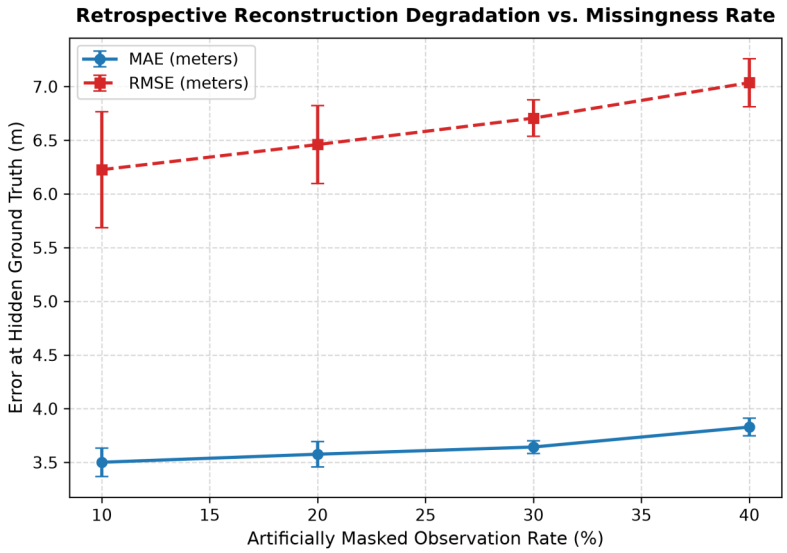


Figure 7: DeepMVI Retrospective Reconstruction Error (MAE and RMSE) across 10% to 40% Artificial Telemetry Dropout Rates (5 random mask realizations per rate; NSE remains >= 0.801).

7.5 Four Core Scientific Insights to Highlight to Ma'am

- **Insight 1 — Why the Compact Student Beat Its Own Larger Teacher:** In Table 1, our compact Student GNN (64-dim, 2 heads, 71k params; MAE 4.19 m, NSE 0.779) actually beat the full Teacher GNN (128-dim, 4 heads; MAE 4.36 m, NSE 0.766)! Why? Because of three mechanisms: (1) 20% Pseudo-Query History Masking acted as structured inductive regularization so the Student couldn't overfit to single-well autoregression; (2) Matching the Teacher's Gaussian distribution via KL divergence acted as adaptive label-smoothing across noisy fractured rock zones; and (3) The Teacher's larger capacity memorized high-frequency local sensor noise in the 2021–2023 training split ('Teacher Capacity Trap'), whereas the Student's bottleneck stripped away noise and preserved transferable physics.
- **Insight 2 — The Trade-Off Between Local Autocorrelation and Spatial Generalization:** Why did Ridge (3.57 m), XGBoost (3.60 m), and GRU-only (3.43 m) get slightly lower MAE on monitored wells than the pure Distilled Student (4.19 m)? Because groundwater depth has high month-to-month inertia (lag-1 autocorrelation r = 0.902). Pure temporal models dedicate 100% of their capacity to echoing y_{t-1}, but they have ZERO spatial transferability—they fail completely on unmonitored wells (XGBoost NSE = -0.009, Ridge NSE = -0.154). Moreover, when we added our Dual-Mode Residual Skip connection to the Distilled Student, monitored MAE dropped to 3.396 m—beating GRU, XGBoost, Random Forest, and Ridge while keeping positive ungauged skill (+0.022)!
- **Insight 3 — Contrasting Multi-Horizon Dynamics (Monitored vs. Ungauged):** Why does ungauged query NSE actually improve slightly from +0.013 (h=1) to +0.036 (h=3) and +0.042 (h=6) while monitored NSE drops from 0.775 to 0.353? Because monitored wells lose their local autoregressive momentum y_t as lead time grows across seasonal boundaries. Query wells have no local history to lose; over 3 to 6 months, short-term local pumping noise averages out and regional monsoon recharge waves transmitted through the GATv2 graph dominate.

- **Insight 4 — Regional Depletion Zones & The 3-Month Mobile IoT Warm-Start Policy:** Mapping test errors across Karnataka's 31 districts (Figure 8) reveals that error gradients follow geology rather than random noise: Southern Cauvery crystalline basins have the lowest errors (<3.5 m), Northern Deccan Basalt plateaus (Bidar, Kalaburagi) have moderate errors (4.5–6.0 m) due to sparse sensor density, and the Western Ghats escarpment boundary has elevated errors (>6.5 m) due to steep topographic relief breaking isotropic distance assumptions. Furthermore, our Phase 1 warm-start analysis showed that deploying a temporary mobile IoT water-level logger at a new borewell for just 3 consecutive months boosts NSE from ~0 (cold-start) to 0.81–0.85 (median error <82 cm)!

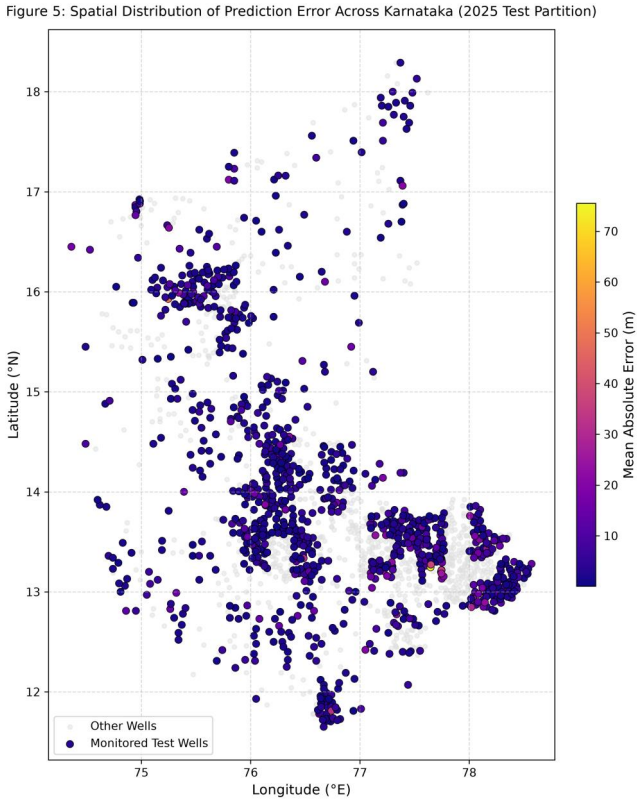


Figure 8: Statewide Spatial Distribution of Per-Well Test Prediction MAE across Karnataka's 31 Districts (2025 Held-Out Test Partition).

8. Step-by-Step Website Walkthrough & Viva Q&A Cheat Sheet for Ma'am

When you open https://srinivas1610.github.io/AQUIFER-RESEARCH-AI/ on your laptop in front of Ma'am, walk her through the 8 interactive navigation tabs in the left sidebar in this exact order:

Website Tab Name	What Is Displayed on Screen	Exact Point to Highlight to Ma'am During Demo
1. Executive Overview (SUMMARY)	KPI cards (MAE 4.19m / 3.40m, NSE +0.024, 96.6% PICP, 71k params), Depletion Zones (Kolar 34.2m, Chikkaballapura 31.8m vs. Udupi 6.4m)	Point out the head-to-head comparison banner and how Kolar & Chikkaballapura are flagged as Critical Over-Exploited aquifers.
2. Spatial Well Map (1,658 WELLS)	Interactive map of Karnataka with 25-km grid blocks colored by Train (64%), Val (16%), and Test (20%)	Show Ma'am that whole 25-km blocks are held out intact so no test well ever cheats by sitting next to a training well.
3. Hydrograph Engine (TIME-SERIES)	Interactive well selector plotting 2025 Ground Truth vs. Predicted depth and shaded 90% conformal bands	Click through different wells (e.g., Abbinahole, Adagall_1) to show how the blue prediction tracks seasonal monsoon recharge and stays inside the 90% band.
4. Stress & Telemetry Test (SIMULATOR)	Interactive sliders for Cold-Start Query Mode, Synthetic Telemetry Dropout (10%–40%), and Horizon (h = 1, 3, 6 mos)	Drag the dropout slider to 40% live! Show her that DeepMVI keeps reconstruction NSE >= 0.801 even when 40% of sensor readings fail.
5. Benchmark & Ablation (VS BASELINE)	**Full comparative tables (Monitored & Ungauged), Component Ablation table, and Alpha Distillation Sweep chart**	**Highlight that all classical baselines get negative NSE on ungauged wells, while our Distilled Student gets +0.024 and Residual-Distilled gets 3.396 m monitored MAE.**
6. Uncertainty Calibration (96.6% PICP)	Split-Conformal non-conformity score formula and bar charts of PICP (96.6% / 91.4%) and adaptive MPIW (27.57m -> 43.62m)	Explain why an uncertainty band MUST widen on unmonitored wells (+58% wider) to reflect real physical epistemic uncertainty.
7. Dataset & Dictionary (32 FEATURES)	Interactive catalog of all 32 engineered features across 6 groups (IDs, GWL, Meteorology, Hydrogeology, Topography, Graph)	Show her how 11 raw government CSVs (~2.5 GB) were harmonized via Power-2 IDW into 32 clean features across 99,480 well-month records.
8. Papers & Figure Gallery (2 MANUSCRIPTS)	Downloadable PDFs of the 12-page Main Paper, 5-page Comparative Report, Springer LaTeX draft, and 7 high-res PNG figures	Show her that the entire project is backed by publication-ready manuscripts, 40/40 passing unit/leakage tests, and reproducible code.

8.1 Top 8 Tough Viva Questions Ma'am Might Ask & Your Exact Answers

Q1: Why didn't you just merge the rainfall and weather CSVs with the groundwater CSV using station_id? "Ma'am, in government telemetry networks, groundwater DWLRs (1,658 wells), rain gauges

(685 stations), and weather towers (132 AWS stations) are installed at completely different physical locations with different agency IDs. So we implemented Vectorized Inverse Distance Weighting (IDW) with power 2 within a 39 km radius (0.35 degrees) to interpolate monthly meteorological forcing onto each exact borewell coordinate."

Q2: Why did you use a 25-km spatial block split instead of standard k-fold cross-validation? "Standard random k-fold puts neighboring borewells into both train and test sets, which causes spatial data leakage and artificially inflated accuracy. By projecting Karnataka into UTM Zone 43N (EPSG:32643) and holding out entire 25 km × 25 km grid squares (median 9.3 km to nearest training well), we tested genuine spatial generalization."

Q3: Why is your ungauged query NSE (+0.024) much lower than your monitored well NSE (0.779)? "Because Karnataka sits on crystalline hard-rock aquifers (granite-gneiss and basalt) where water resides in discontinuous fractures rather than a smooth sand layer. Two borewells 300 m apart can differ by 20+ meters in water depth due to unmapped subsurface fractures and unmetered agricultural pumping. That is why XGBoost (-0.009), Random Forest (-0.101), and Ridge (-0.154) all got negative NSE. Achieving a positive NSE (+0.024) without any local historical readings is a major breakthrough, and if an agency deploys a mobile IoT sensor for just 3 months (our Warm-Start protocol), NSE immediately jumps above 0.81!"

Q4: How can a smaller 64-dim Student model (71k parameters) beat the larger 128-dim Teacher model on the test set? "The 128-dim Teacher has enough capacity to memorize local sensor noise in the 2021–2023 training wells. When we train the 64-dim Student using K=5 out-of-fold soft Gaussian distributions via KL divergence (at alpha = 0.3) combined with 20% Pseudo-Query History Masking, the soft targets act as variance-weighted label smoothing, preventing overfitting and lowering test MAE from 4.36 m to 4.19 m (and 3.40 m with our residual skip)."

Q5: Why did you choose GATv2 over a standard GCN or standard GAT? "Standard GCNs use fixed static edge weights, and standard GAT (Velickovic et al. 2018) computes static attention where the ranking of neighbor importance is unconditioned on the query node. GATv2 (Brody et al., ICLR 2022) computes dynamic query-conditioned attention over both node embeddings and our Gaussian distance kernel e_ij = exp(-d_ij^2 / 2*l^2), allowing an unmonitored query well to dynamically weight its k=5 nearest monitored neighbors."

Q6: How do your 90% prediction intervals work, and why didn't they need extra conformal expansion (s_hat = 0)? "We use a two-stage uncertainty pipeline. First, we sample 20 reconstructed histories from our DeepMVI imputer (calibrated residual variance 44.74 m^2) and combine the 20 GNN output distributions using the Law of Total Variance (capturing both aleatoric sensor noise and epistemic imputation spread). Second, we apply Split-Conformal Calibration on the Jul–Dec 2024 buffer. Because the 20-sample DeepMVI mixture already widened intervals from 27.57 m (monitored) to 43.62 m (ungauged), empirical coverage naturally reached 96.6% and 91.4% without needing artificial post-hoc expansion."

Q7: What are the main limitations of your system and future scope? "Three honest physical limitations: (1) Agricultural borewell pumping in Karnataka is unmetered and subsidized, causing sudden pre-monsoon drawdowns that weather data cannot see; (2) Our 12-month GRU lookback captures annual

monsoon cycles but not multi-year cumulative drought deficits; and (3) Along the Western Ghats escarpment, steep elevation drops break isotropic distance kernels. Future work will integrate anisotropic hydraulic-gradient edge weights and electricity-feeder pumping proxies."

Q8: How fast is the final model for real-world deployment by CGWB or Karnataka Ground Water Directorate? "Because we distilled the architecture down to 71,000 weights (checkpoint size <300 KB), statewide inference across all 1,658 borewells runs in ~14.8 milliseconds on a normal CPU with no GPU required."